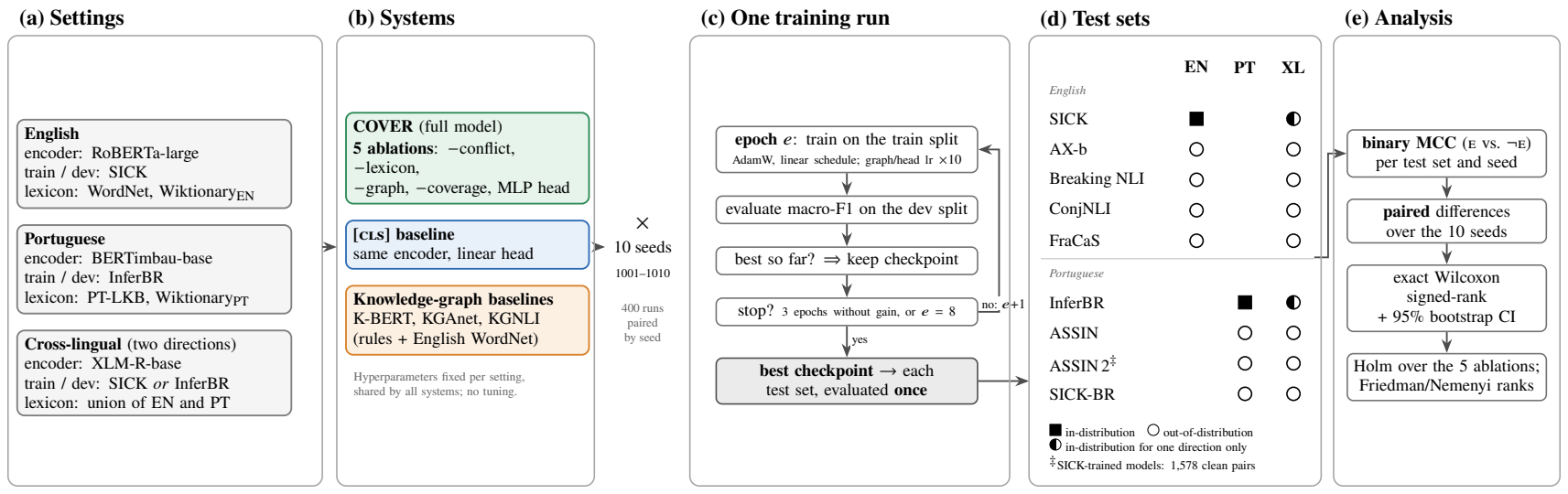

(a) Settings
English
encoder: RoBERTa-large
train / dev: SICK
lexicon: WordNet, Wiktionary$_{EN}$
Portuguese
encoder: BERTimbau-base
train / dev: InferBR
lexicon: PT-LKB, Wiktionary$_{PT}$
Cross-lingual (two directions)
encoder: XLM-R-base
train / dev: SICK or InferBR
lexicon: union of EN and PT
(b) Systems
COVER (full model)
5 ablations: −conflict, −lexicon, −graph, −coverage, MLP head
[CLS] baseline
same encoder, linear head
Knowledge-graph baselines
K-BERT, KGAnet, KGNLI (rules + English WordNet)
Hyperparameters fixed per setting, shared by all systems; no tuning.
×
10 seeds
1001–1010
400 runs paired by seed
(c) One training run
epoch e: train on the train split
AdamW, linear schedule; graph/head lr ×10
evaluate macro-F1 on the dev split
best so far? ⇒ keep checkpoint
stop? 3 epochs without gain, or e = 8
no: e+1
yes
best checkpoint → each test set, evaluated once
(d) Test sets
EN
PT
XL
English
SICK
AX-b
Breaking NLI
ConjNLI
FraCaS
Portuguese
InferBR
ASSIN
ASSIN 2‡
SICK-BR
in-distribution
out-of-distribution
in-distribution for one direction only
‡SICK-trained models: 1,578 clean pairs
(e) Analysis
binary MCC (E vs. ¬E) per test set and seed
paired differences over the 10 seeds
exact Wilcoxon signed-rank + 95% bootstrap CI
Holm over the 5 ablations; Friedman/Nemenyi ranks